Kyeongho Kim

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Profile

I am an undergraduate student in Statistics at Inha University, working under the supervision of *Prof. Ilsang Ohn* in the Statistical Theory Research Lab and *Prof. Doowon Choi* in the MIND Lab. (Modeling and Interpreting Data). My research interests include Reliable & Interpretable Machine Learning, Statistical Perspectives on Neural Networks (Theory & Applications).

Education

Inha University Mar. 2021 – Feb. 2027 (Expected)

B.S. in Statistics, Double Major in Software Convergence Engineering

Completed in 7 semesters, First–class honors

GPA: 4.32 / 4.50 (Major: 4.42 / 4.50)

Selected Courses

Math & Statistics : Mathematics for Artificial Intelligence I (A+), II (P), Bayesian statistics (A+), Multivariate Statistics and practice (A+), Statistical Machine Learning (A+), Computational Statistics (A+), Survival Analysis and Lab (A+), Time Series Analysis (A+), Categorical Data Analysis (A+), Mathematical Statistics (A+), Regression Analysis (A+), Linear Algebra (A+), Social Research Methodology (A+)

Computer Science : Computer Network (A+), Algorithm (A+), Object-Oriented Programming, Statistical Software and Practice (A+), Programming Language (A+)

Publications

Rate-Optimal Neural Boundary Detection from Unlabeled Noisy Images

Kyeongho Kim and Ilsang Ohn, *under review*, **arXiv**

A Partial-Correlation-Refined Attention Framework for Localized Feature–Time Explanation in Manufacturing Quality Diagnosis

Kyeongho Kim, Sangjin Choi and Doowon Choi. *under review.*

Research Experience

Statistical Theory research Lab. Sep. 2025 – present

Undergraduate Researcher (Advised by Prof. Ilsang Ohn)

- Developed neural network-based methods for unsupervised boundary detection in noisy intensity images, using surrogate loss functions and analyzing rate-optimal convergence, under varying signal-to-noise ratios.
- Researched Discrete Diffusion models to remove noise from weak signal images, researching and performing not only gaussian diffusion, but also poisson diffusion (for poisson synthetic image).

MIND Lab. (Modeling and INterpreting Data) Jun. 2025 – Aug. 2026

Undergraduate Researcher (Advised by Prof. Doowon Choi)

- Developed an interpretable anomaly detection framework for multivariate time-series data, focusing on Partial Correlated Attention, an extension of Transformer-based attention for feature-time importance analysis and root-cause identification.
- Researched a more flexible VAE-based modeling framework by combining variational autoencoders with symbolic regression or online learning appraoch

UNIGENCE Co., Ltd. Aug. 2025 – Sep. 2025

Algorithm development Internship

- Developed model architectures and pattern identification algorithms for industrial sensor signals, with a focus on fault detection and time-series anomaly analysis, which is conducted Grad-Cam, XCM, Shapley value.

Honors & Awards

Seminar in Inha Statistic Graduate School – Third Prize Feb. 2026 – April. 2026
Presented a project on boundary detection from noisy images, focusing on boundary recovery under varying noise levels and statistical convergence rate of the proposed model. Awarded Third Prize at the Inha Statistics and Data Science Graduate School seminar. **[Slides]** **[Conference document]**

KIIE Undergraduate Project Competition – Bronze Prize Sep. 2025 – Nov. 2025
Conducted an interpretable attention module for time-series analysis framework to an industrial engineering problem, focusing on practical anomaly detection and root-cause identification. The project was recognized with the Bronze Prize at a competition hosted by the Korean Institute of Industrial Engineers (KIIE). **[Slides]**

Image Classification with Vision Transformer Apr. 2025 – Jun. 2025
Performed landmark image classification challenge using Vision Transformer (ViT), achieving 2nd level accuracy in a Mathematics for Artificial Intelligence I course.

Projects

Conformalized Survival Analysis May. 2026 – Jun. 2026
Reviewed and studied conformalized survival analysis for general right-censored data, and derived lower prediction bounds for censored patient survival data in the Survival Analysis and Lab course. **[Slides]**

Korea Meteorological Administration Weather BigData Contest May. 2026 – Jun. 2026
Developed a LightGBM-based multiclass prediction pipeline on 2.4M cattle records by integrating livestock, weather, farm, pedigree, and genetic data. Engineered THI-based heat-stress exposure features and analyzed feature importance to identify production-related factors associated with slaughter grades.

Data-Driven Customer Segmentation and Income Conformal Prediction May. 2025 – Jun. 2025
Performed customer segmentation using K-means on high-dimensional consumption and demographic data, followed by Random Forest–based segment classification with conformal prediction. Conducted income-level prediction from consumption items using PCA to address multicollinearity and uncertainty-aware prediction, enabling strategic extrapolation to new customers in Multivariate Statistics and Practice course.

Extra-Curricular Activities

President, Student Council Jan. 2025 – Dec. 2025

- Led Inha statistics student council, developed activities and coordinated academic and administrative initiatives.

Teaching Assistant Sep. 2025 – Present

- Serving Inha statistics TA for serveral lectures (Mathematical Statistics, Regression Analysis, Statistical Software and Practice)

Learn-Stat. - Mentor Dec. 2025 - present

- Mentor of Learn-Stat. : Student conference in the Department of Statistics of Inha University

Statistics Mentoring - Mentor Mar. 2025 - Dec. 2025

- Mentor of Statistics Mentoring: Statistics Mentoring for students in Statistics of Inha University

High school Math Instructor Jan. 2024 – present

- Teaching mathematics to high school student in academy

Army Accomodation Medal (R.O.K Army) Jul. 2022 – Jan. 2024

- Completed Full Military Service as Sergeant